A method for breast mass segmentation using image augmentation with SAM and Receptive Field Expansion

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ABSTRACT

With the development of deep learning methods and their wide application in medical image segmentation tasks, mammography, used for early breast cancer screening, can further assist clinicians in diagnosis to a certain extent. Due to the loss of fine-grained features in the downsampling process and the very few available mammograms, the main methods for extracting masses from the whole mammography are relatively ineffective. In this paper, we propose a novel mass segmentation model for mammograms based on two U-Net network models, and integrate the Receptive Field Block (RFB) module in-between to enhance the deep features captured by the model. We augmented the input image using Segment Anything Model (SAM) and evaluated the proposed architecture on two public datasets, namely the Curated Breast Imaging Subset of Digital Database for Screening Mammography (CBIS-DDSM) and INBreast, as well as on a private dataset. The results show that the proposed model for segmenting masses in ROI regions can achieve high Dice scores of 92.18% and 89.85%, and Intersection over Union (IoU) scores of 85.47% and 80.80% on both INBreast and the private dataset. In addition, our model for segmenting masses on whole mammographs in CBIS-DDSM dataset can achieve Dice scores of 58.10% and 41.96% IoU scores. Besides, our model performance improved to Dice score 61.78% and IoU score 43.00% on whole mammographs in private dataset using SAM-augmented input images.

CCS CONCEPTS

• **Computing methodologies** → *Image segmentation.*

KEYWORDS

U-Net, Mammography, RFB, SAM, semantic segmentation

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1 INTRODUCTION

Breast cancer remains a major cause of suffering and death among women globally[1]. In 2022, breast cancer was the most striking cancer in the United States, and one of the five most diagnosed cancer in China [2]. More particularly in China, the situation is alarming: the trend of the breast cancer incidence has been the highest over 20 years and is becoming the second type of cancer causing death after lung cancer [3]. When the patients can survive it, the later the breast tumor is detected, the longer, more exhaustive, damaging and emotionally-changing the healing process can be for women [4]. In this alarming context, automatic medical image processing can play a vital role in diagnosing and treating it as early as possible. Among the existing breast cancer screening tests, mammography captures an X-ray image of the breast, or mammograms, and remains the most efficient test to find early breast cancers. Abnormal masses in breast mammograms can indicate the stage of cancer to some extent[5], and their automatic recognition can valuably assist physicians in determining the type of tumor and aid in early screening and diagnosis of breast cancer.

In recent years, deep learning [6] methods have shown great performance in medical image segmentation, where most segmentation models rely on being encoder-decoder networks, represented

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by the U-Net architecture [7]. Compared to traditional segmentation approaches, encoder-decoder convolutional neural network (CNN) architectures achieve superior performance in breast mass segmentation tasks due to their use of skip connections, which fuse low-level and high-level semantic features to produce the final mask. However, current deep learning architectures set receptive fields of the same size across feature maps, which may lead to some information loss in features. More precisely, many improved architectures from the U-Net structure focus only on improving the deep feature extraction algorithm, ignoring the loss of intermediate-to high-resolution features during the first downsampling process of the encoder.

This paper relies on the Connected-UNet [8] as basic structure and adopts two strategies from there. Firstly, to tackle the problem of feature information loss, the original Atrous Spatial Pyramid Pooling (ASPP) block from the Connected-UNet is replaced by the Receptive Field Block (RFB) [18]. RFB implements various kernel sizes in each convolutional branch of its multi-branch structure and adds up the complementary output features together, expanding significantly the receptive field. Moreover, the recently released Primary Feature Conservation (PFC) block [9] replaces the first encoder block in the Connected-UNet structure, which captures rich features from the input mammograpy image, compensate for fine-grained features loss in the first downsampling block and increases further the receptive field size. Secondly, we leverage the recent Segment Anything Model (SAM) [12] to extract a prior segmentation map. This prior map segments the breast region in a mammogram and are used jointly with this mammogram as input to our model, which adds more data expressiveness, can guide the model in segmenting tumors within the breast region and ultimately improves the segmentation results of our model.

We used two public datasets, CBIS-DDSM [10] and INbreast [11], as well as a private dataset to evaluate the model's performance in extracting masses in the region of interest (ROI) and whole mammograms, separately.

Our main contributions can be summarized as follows:

(1) We propose to integrate the Receptive Field Block (RFB) into the Connected-UNet structure. Replacing the former ASPP blocks, its design particularly tackles the problem of low segmentation accuracy due to restricted receptive fields, and can effectively compensate for the lack of fine-grained details, inherent to the U-Net network.

(2) We add the recently designed Primary Feature Conservation (PFC) block at the beginning of the improved Connected-UNet structure. In place of a basic encoder block, the PFC block extracts primary features from the input mammograpy image and increase the network receptive field, shrinking further the information loss in features.

(3) We adopt the recently developed large model, named Segment Anything Model (SAM) for input augmentation in the employed dataset. This SAM module provides a prior feature map for downstream segmentation tasks. To the best of our knowledge, this paper is the first attempt to use the SAM model for input augmentation for the task of mass segmentation in mammograms.

2 RELATED WORKS

In the last years, deep learning methods have performed well in the field of medical image segmentation and have shown great potential in assisting clinicians in their diagnostic work, especially in the tasks of mass detection and segmentation of mammography images. Ronneberger et al. [7] proposed the U-Net semantic segmentation network, where skip connections between the encoder and decoder can transfer the information between the shallow and deep outputs of the network, which demonstrated good performance in a variety of medical image segmentation tasks. Numerous enhanced architectures prioritize the enhancement of algorithmic deep feature extraction, with different scale features having already demonstrated their potential in the domain of medical image segmentation. Lou et al. [13] designed efficient dual channel efficient CNN architecture to replace encoder and decoder in a standard U-Net network, which can overcome the problem of insufficient spatial features. Sun et al. [14] proposed a novel attention-guided dense-upsampling network. It compensates the information loss of bilinear upsampling by dense upsampling. Li et al. [15] integrated the merits of residual learning and probabilistic graphical modelling with standard U-Net, and proposed Conditional Residual U-Net (CRU-Net), to improve the U-Net segmentation performance. Accordingly, Zhu et al. [16] proposed a multi-scale FCN model followed by a conditional random field (CRF) for mammographic mass segmentation. Another work proposed by Singh et al. [17] developed a conditional Generative Adversarial Network (cGAN) [19] for breast tumor segmentation. These models perform well, but they lose certain fine-grained features in the downsampling process, resulting in unclear boundaries for mass segmentation, and there are still difficulties in extracting small areas of mass regions in high-resolution medical images.

Moreover, there are many works proposing cascaded U-Net architecture for medical image segmentation. Jha et al. [20] proposed to use a double U-Net architecture called Double U-Net for medical image segmentation tasks, which takes two encoders and two decoders along with ASPP [21] module: the first U-Net uses a pretrained VGG-19 as the encoder, which has already learned features from ImageNet and another U-Net at the bottom. Das et al. [22] proposed a model called Contour-Aware Residual W-Net which consists of two U-Nets, the first U-Net predicts the boundaries of the objects and the second one produces accurate segmentation maps based on it.

In addition, the utilization of data enhancement techniques is prevalent in medical image segmentation endeavors owing to the limited availability of reliable data within medical images. The mainstream data enhancement methods often use hand-designed transformations, such as rotation, cropping. Moreover, many studies have focused on increasing the size of dataset by adding synthetic data generated using an image-to-image translation method, e.g., Cycle-Consistent Generative Adversarial Network (CycleGAN) [23], between the different mammography data sets. These image enhancement methods are usually used in low-level, noise-containing situations, usually to recover and reconstruct the image, without considering image enhancement in terms of high-level structures. Due to the introduction of the SAM model, many works have applied the SAM model to various medical image segmentation tasks

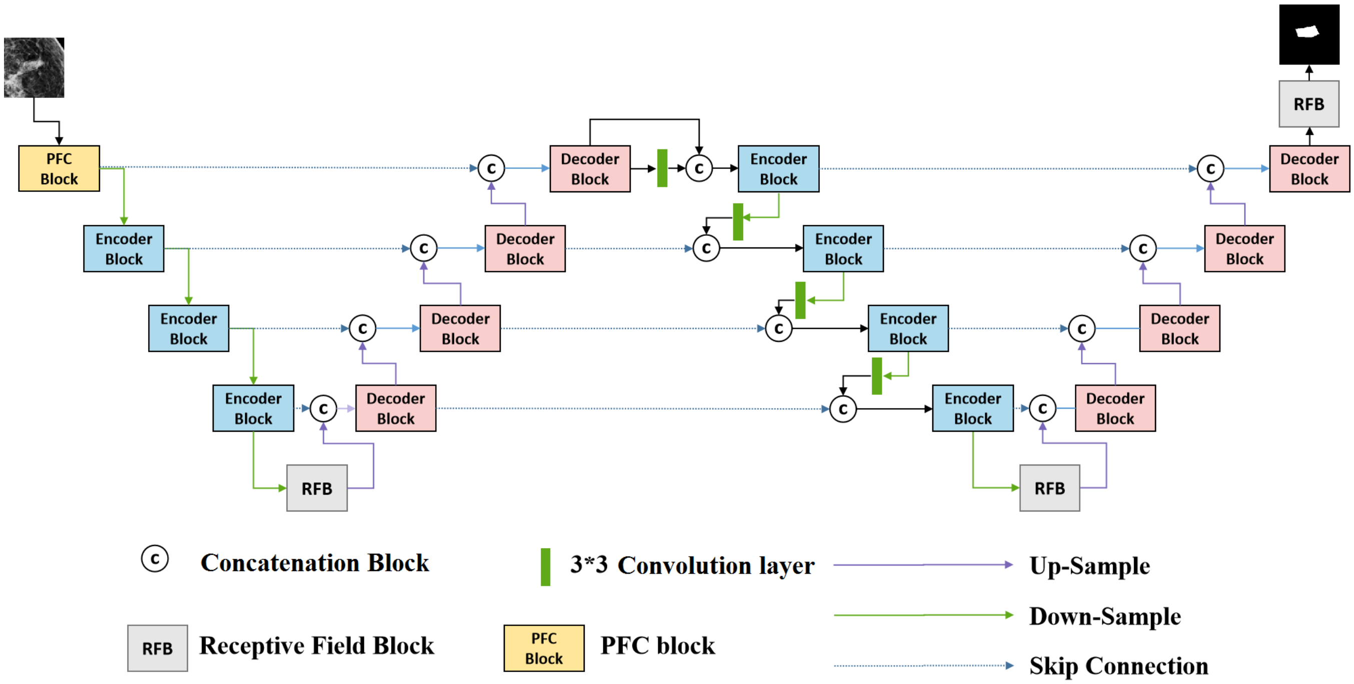

Figure 1: Our proposed network architecture. Receptive Field Blocks (RFB) replace ASPP blocks in the original Connected-UNet for reducing the feature information loss in key stages, whereas the PFC block expands the receptive field at the first encoding stage without increasing the computational cost or number of parameters.

[24], [25], [26], [27]. which usually do not produce satisfactory segmentation results. In our work, we combine the original mammography images in the segmentation output of the SAM model and employ the SAM model for input augmentation, in the view of constructing a downstream mammography mass segmentation model.

In this paper, we propose a cascaded encoder-decoder architecture as a solution to the limitations observed in mainstream networks. Our approach involves the implementation of PFC to incorporate remote spatial features at the low semantic layer. Additionally, to overcome the challenge posed by the limited number of mammography images, we employ the SAM model for input augmentation. This augmentation technique is applied to three distinct mammography image datasets, resulting in improved segmentation outcomes for our model.

3 METHODOLOGY

In this section, the proposed network architecture and the whole flowchart of the SAM-based input augmentation will be presented, finally we will give the evaluation metrics.

3.1 Model architecture

In the U-Net network and its derived variants, the problems of limited receptive fields and feature information loss tends to limit their own segmentation accuracy. In this paper, we propose to expand the receptive field of a cascaded U-Net model architecture, specifically Connected-UNet [8], shown in Figure 1. In the Connected-UNet, the features from the encoders are linked directly to the features of their corresponding scale in the decoders through skip connections. We chose particularly the Connected-UNet as base network for its skip connections between the decoder low-level and high-level semantic features of its first U-Net and the corresponding encoder features of the second U-Net, which can further compensate for the fine-grained features lost during the downsampling process.

To expand the receptive field, the Receptive Field Block (RFB)[18] module is employed between the encoder and decoder of each U-Net separately, and at the end of the whole model, replacing the former ASPP module [8]. The RFB is a multi-branch convolution block with convolutions of different kernel sizes and dilation expansion rates. Compared with ASPP, the expansion rates used in RFB does not exceed 5 and do not have a common divisor greater than 1, which prevents the appearance of gridding effect in feature maps. In addition, RFB applies additional convolution layers before the dilated convolutions in each branch, where the branch of the 3×3 convolution with an expansion rate of 5 offers a receptive field of 11×11, which is reasonable more than the ASPP module, considering the proportion of the breast masses in the target mammograms. After combining the expanded features with a shortcut connection, as depicted in Figure 2, RFB extracts richer information to locate the breast mass more clearly.

In addition, in order to encourage the segmentation network to process rich features from mammograms, a PFC module is used as at the beginning of the first U-Net, replacing its first downsampling block. PFC employs a depth-separable convolution consisting of a

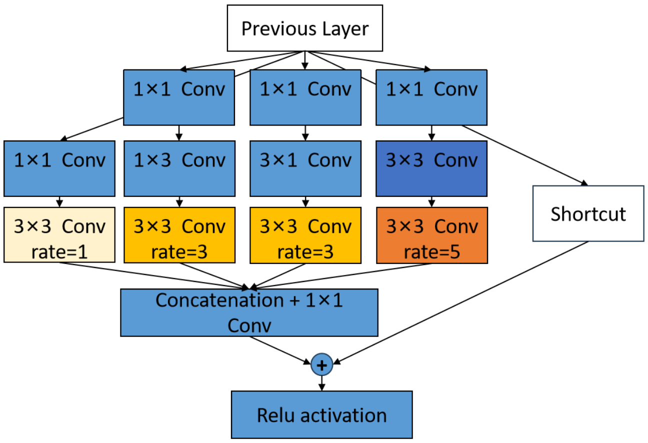

Figure 2: Receptive Field Block (RFB) structure

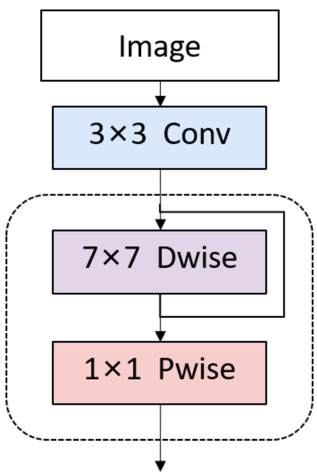

Figure 3: PFC block structure

7×7 depthwise convolution followed by a 1×1 pointwise convolution, as illustrated in Figure 3. A 3×3 standard convolution is added to module head to keep a similar structure with the replaced downsampling encoder of two 3×3 convolution layers. Compared to a 7×7 standard convolution, separating the depthwise convolution and the pointwise convolution reduces computational cost and parameters, which supports the application of a large kernel size to the depthwise separable convolution to offer larger receptive field, extract richer features from mammograms and merge long-range spatial information at low semantic level. Each convolution layer is associated with a ReLU activation function and batch normalisation.

In sum, the PFC module processes a 3×3 standard convolution for higher feature extraction performance, followed by a 7×7 depthwise separable convolution for capturing longer-range features efficiently. Added into our Connected-UNet architecture, the PFC can further improve the segmentation performance without increasing the number of parameters or computational cost.

3.2 Input Augmentation using SAM

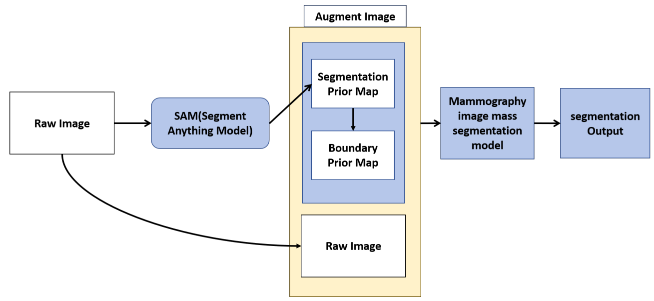

Figure 4: Input Augmentation with SAM is used to improve mammogram mass segmentation.

In this section, we will describe how we use the segmentation prior map generated by the SAM model, the boundary priori map, and then enhance the input image using them. The entire flowchart is shown in Figure 4.

In the grid prompt setting, SAM uses a grid prompt to generate segmentation masks for an input mammography image. The segmented masks generated by SAM are then stored in a list. For each segmentation mask in the list, we draw the mask on a new segmentation prior map using the value suggested by the mask's stability score. In this paper, we also generate a boundary prior map according to the masks generated by SAM. We draw the exterior boundary of each segmentation mask from the list and set all the boundaries to generate a boundary prior map. The mammography image mass segmentation task can be considered as a three-class segmentation task, where the first class corresponds to the background, the second class corresponds to the ROI, and the third class corresponds to the boundary. After obtaining this segmentation prior map, we add its ROI prior map as a second channel of the raw mammography image, and the boundary prior map to a third channel of the raw mammography image. Thus the SAM-enhanced input image is obtained. The input image augmentation with SAM can improve image segmentation performance, visualised in Figures 5, 6 and 8.

3.3 Evaluation metrics

In the context of mass segmentation in mammography images, it is observed that the abnormal mass constitutes a relatively small fraction of the overall image. In this regard, the conventional segmentation evaluation metrics, namely precision and recall, are considered unsuitable. To aptly assess the effectiveness of our model architecture, this paper employs two other commonly-used evaluation metrics in the field of semantic segmentation for our experimental analysis: the Dice similarity score, or F1 score, and the IoU score, or Jaccard score. Their formulations are given in Eq. 1 and 2:

$$Dice\ score(A, B) = \frac{2 * (A \cap B)}{2(A + B)} \tag{1}$$

$$IoU\ score(A, B) = \frac{A \cap B}{A \cup B} \tag{2}$$

Table 1: Details of the mammogram datasets employed in our experiments. From left to right, the columns correspond to the number of raw mammography images, method for extracting of mass-centered ROIs from raw mammograms, number of mass-centered ROIs images, and distribution of the training set/test set, respectively. Note: the number of ROI images obtained by cropping is more than the number of whole images due to the possibility of multiple masses in a single image.

Dataset	RAW MGs data	Method for extracting mass-centered ROIs	Raw ROIs data	Training data 70%	Testing data 30%
CBIS-DDSM	1568	Get the maximum outer matrix of mass with real labels	1798	1389	409
INBreast	107	Get the coordinates of the mass bounding box given on the INbreast official website	107	87	20
Private	348	Get the maximum outer matrix of mass with real labels	348	278	70

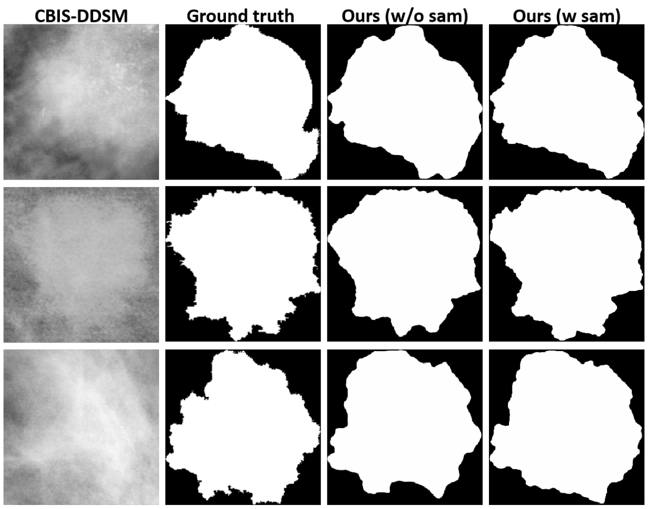

Figure 5: Visual segmentation results of our network architecture on the cropped ROIs from CBIS-DDSM dataset, with and without SAM-augmented input images.

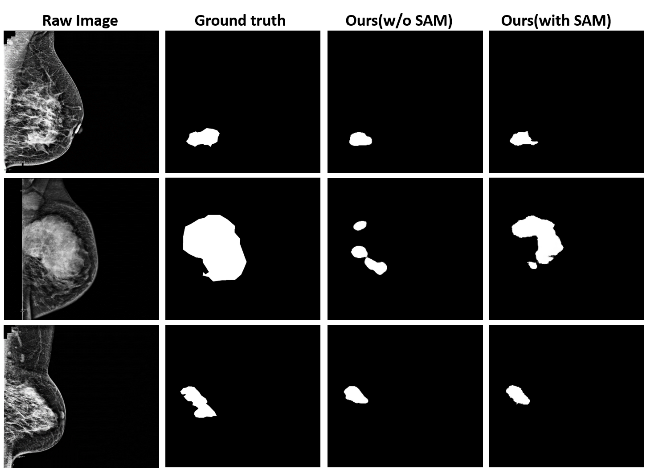

Figure 6: Visual segmentation results of our network architecture on the private dataset, with and without SAM-augmented input images.

4 EXPERIMENTAL RESULTS & DISCUSSION

In order to assess the performance of the proposed model architecture for breast mass segmentation in mammography images, we conducted experiments on mass-centered image patches (i.e., ROI) as well as on the whole mammograms. The employed mammogram datasets include two publicly available datasets, namely CBIS-DDSM and INBreast, and a private dataset. These datasets are described in section, and their details are summarized in Table 1.

In this paper, image processing was performed on the raw medical images to obtain higher quality mammograms, which includes format conversion, min-max normalisation, artefact removal operations, breast orientation standardisation in mammograms, and contrast-constrained adaptive histogram equalisation filtering.

4.1 CBIS-DDSM

CBIS-DDSM is a publicly available digital mammography image dataset that contains 2620 mammography images, including 1168 cases of breast mass and their corresponding ground truth annotations. This paper uses 1798 of the total 2620 images for evaluating our model performance in mass segmentation. The CBIS-DDSM mammograms may contain single or multiple breast masses.

In order to obtain the corresponding mass-centered patch dataset, we cropped 1798 ROIs containing the mass according to the mass location and contour in the segmentation mask marked by the professional doctors. We then divided the training set and test set according to the ratio of 8:2 to ensure that the mass part and the background part in the region of interest are both present. In this paper, a total of 1389 regions of interest containing masses extracted from 1214 mammography images in CBIS-DDSM are used as the training set, and the other 409 regions of interest containing masses extracted from 354 mammography images are used as the test set. All images were resized to 256 × 256.

We conducted experiments on the CBIS-DDSM patch dataset to compare qualitatively our proposed model with similar models. Figure 7 depicts an example of their segmentation results, where our model yields more fine-grained segmentation.

Similarly, we carried out qualitative experiments on CBIS-DDSM whole mammograms, when using the SAM augmented image as input. Figure 8 shows an example of the segmentation results, where our model can better segment masses in mammograms that contain one or even multiple masses.

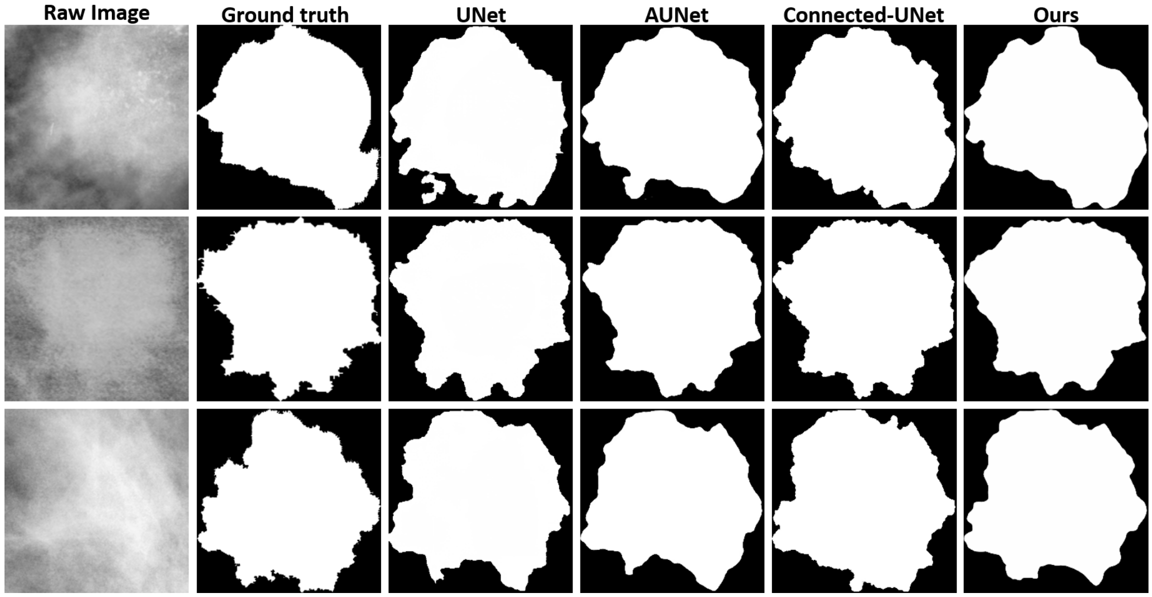

Figure 7: Visual segmentation results of different networks on the cropped ROIs from CBIS-DDSM dataset. From left to right, the columns correspond to the input original images, the ground truth segmentation labels, the segmentation results of UNet, AUNet, Connected-UNet and our proposed model, respectively.

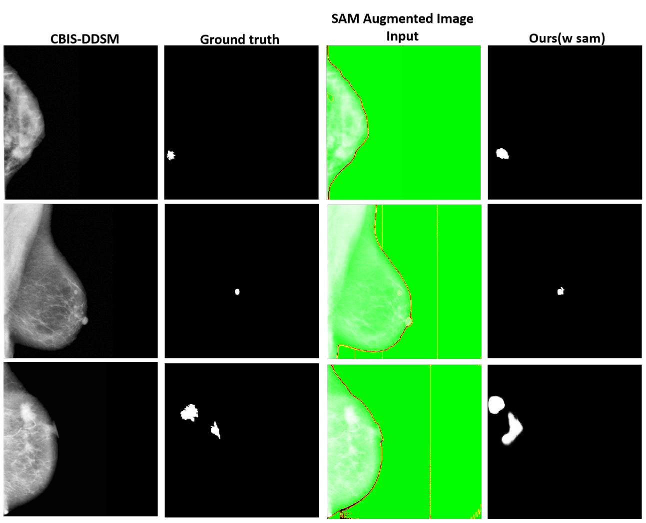

Figure 8: Visual ablation experiment results of the prior segmentation map from SAM and our model using the original mammogram and this prior map as input, on CBIS-DDSM dataset. From left to right, the columns correspond to the input original whole mammogram image, the ground truth labels, the SAM-augmented image input, and the final segmentation results.

4.2 INBreast

INBreast is a publicly available mammography dataset that contains 410 mammogram images from 115 patients with a variety of breast lesions such as masses, calcifications, asymmetries, and structural variants. In this paper, 106 mammogram images, containing masses and their corresponding real annotations, are used to test our model performance on the mass segmentation task. Similarly to CBIS-DDSM dataset, an INBreast patch dataset is built.

We conducted experiments on INBreast patch dataset and compared our model with other similar models. From Table 2, we can see that our proposed architecture achieves the best performance results in segmentation mass on INBreast patch dataset. The results further show that the proposed model for segmenting masses in the ROI region can achieve a high Dice score of 92.18% and an IoU score of 85.47%.

4.3 Private dataset

Moreover, this paper exploits a privated dataset of 348 mammography images of size 1024 × 1024. This dataset consists of mammograms containing breast masses and their corresponding binary mass masks annotated by radiologists in conjunction with biopsy reports. Following a ratio of 8:2, 278 of these mammograms are for the training set and the remaining 70 mammograms are for the validation set. Similarly to the publicly available dataset, a private patch dataset is built from the ROI around breast masses. Table 3 and Table 4 list the performance of our proposed model compared to similar models, respectively on the private patch dataset and the whole mammograms of the private dataset.

From these tables, we can see that our proposed architecture has the best performance for mass segmentation in patches, as well as for mass segmentation in whole mammograms, when using the SAM augmented image as input. More specifically, the proposed model for mass segmentation in the ROI regions can achieve a

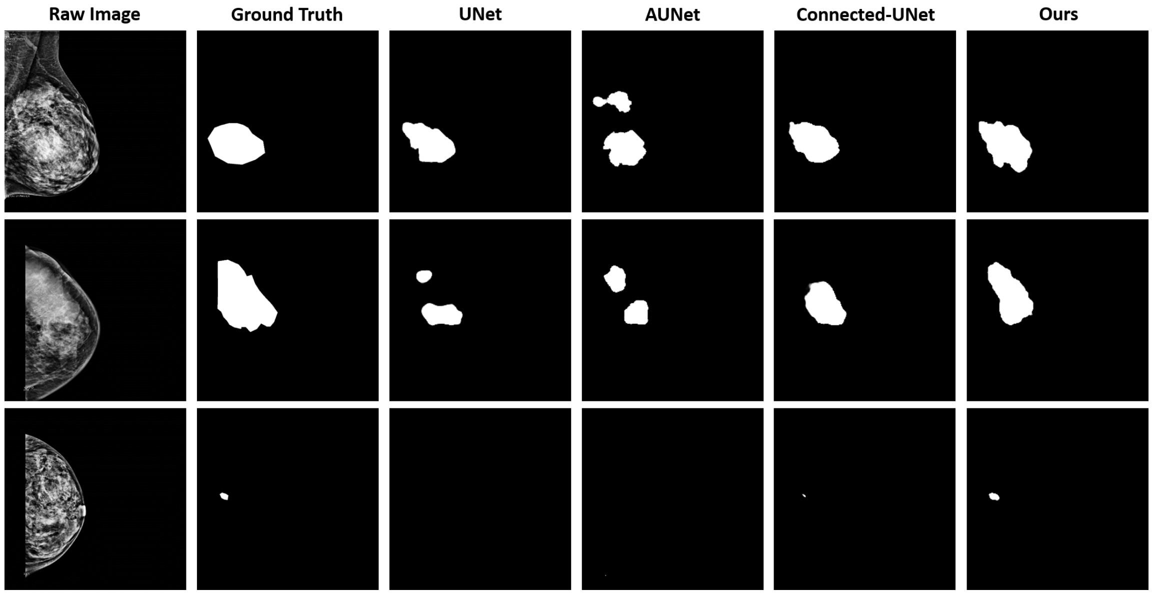

Figure 9: Visual segmentation results of different networks on the entire mammographs from the private dataset. From left to right, the columns correspond to the input original images, the ground truth segmentation labels, the segmentation results of UNet, AUNet, Connected-UNet and our proposed model, respectively.

Table 2: Segmentation performance comparison of our proposed model with state-of-the-art methods on INBreast dataset. Bold values indicate best results.

Model	Dice score	IoU score
Attention UNet [14]	91.92 %	85.06%
Connected-ResUNet [8]	91.42%	84.12%
FCBFormer [28]	89.81%	71.51%
FCN-CRF [29]	90.97%	72.30%
Ours	**92.18%**	**85.47%**

Table 3: Segmentation performance comparison of our proposed model with state-of-the-art methods on the private patch dataset. Bold values indicate best results.

Model	Dice score	IoU score
Attention UNet [14]	89.42%	80.23%
ResUNet [30]	89.09%	80.03%
FCBFormer [28]	80.47%	68.32%
MSNet [31]	87.55%	79.66%
Ours	**89.85%**	**80.80%**

high Dice score of 89.85%, and an IoU score of 80.80% in the ROI regions. In addition, the proposed model for mass segmentation, only considering whole mammograms as input, can achieve a high Dice score of 58.10%, and an IoU score of 41.96%. Besides, when using the SAM augmented image as input, the results improved to a Dice score of 61.78% and an IoU score of 43.00%.

Table 4: Segmentation performance comparison of our proposed model with state-of-the-art methods on the whole mammograms from the private dataset. Bold and italic figures indicate best and second best results, respectively.

Model	with SAM	Dice score	IoU score
U-Net [7]		49.66%	35.38%
	√	52.63%	38.71%
A-UNet [14]		53.43%	39.98%
	√	54.86%	39.76%
Connected-UNet [8]		59.45%	**43.20%**
	√	*60.20%*	41.85%
Ours		58.10%	41.96%
	√	**61.78%**	*43.00%*

CONCLUSION

In this paper, we propose a new method for mammography mass segmentation, that combines features of different levels across two cascaded U-Net models. In order to improve model segmentation performance, we also augmented the input images using the SAM model, which is assessed on two major public datasets, namely CBIS-DDSM, INbreast, and our private dataset collected from a local hospital. The experiments on these three datasets, separately on ROI regions and whole images, proved the effectiveness of our model. Despite the effectiveness of its design, our model still has room for improvement in different aspects. In our future work, we plan to conduct additional experiments to determine the ideal choice of hyperparameters for our model. In addition, we plan to evaluate the generalization ability of the proposed model on other related medical image segmentation tasks.

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